

RS002 Week 14 — English Worksheet

Topic: Best-of-3 Tournament + Type Advantage · **Course:** Pokédex App · **Time:** about 45 minutes

This week you think about a **best-of-3 tournament**. You read how wins and losses are tracked in a `while` loop, and how a **type-advantage** bonus uses a small dictionary (water beats fire, and so on). Your job is to read the code closely, debug it on paper, and explain how it works in your own words.

Words to know: **tournament**, **while loop**, **type advantage**, **get**, **best of three**

1 · Recall

From memory:

Write the line that picks a random Pokémon dictionary from a list called `pokedex` (assume `random` is imported).

2 · Predict 🧠

Read each block. Write what you think it does — no need to run it.

```
TYPE_ADVANTAGE = {"물": "불", "불": "꿀"}
print(TYPE_ADVANTAGE.get("물"))
print(TYPE_ADVANTAGE.get("전기"))
```

.get returns the value, or **None** if the key is missing. What two lines print?


```
wins = 0
losses = 0
while wins < 2 and losses < 2:
    wins = wins + 1
    print(f"{wins}승 {losses}패")
```

The loop keeps going while BOTH are under 2. Write every line it prints, and say when it stops.


```
attacker = {"type": "물"}
defender = {"type": "불"}
TYPE_ADVANTAGE = {"물": "불"}

if TYPE_ADVANTAGE.get(attacker["type"]) == defender["type"]:
    print("효과적!")
else:
    print("보통")
```

Does **효과적!** print? Walk through the comparison.

3 · Spot the Bug 🐛

Read what each block is **supposed** to do, fix it on paper, then explain the fix.

Bug A — This loop should stop once **either** side reaches 2. Right now it stops only when both reach 2, which can loop forever.

```
wins = 0
losses = 0
while wins < 2 or losses < 2:
    pass # battle happens here
```

Hint: swap `or` for `and`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The advantage check should be safe even when the type is not in the table. Right now an unlisted type crashes with a `KeyError`.

```
TYPE_ADVANTAGE = {"물": "불", "불": "폭"}
attacker = {"type": "전기"}
defender = {"type": "물"}

if TYPE_ADVANTAGE[attacker["type"]] == defender["type"]:
    print("효과적!")
```

Hint: use `.get(...)` instead of `[...]` so a missing key returns `None`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — A win should add **one** to `wins`. Right now it resets `wins` to 1 every round, so the trainer can never reach 2.

```
wins = 0
# round 1 won
wins = 1
# round 2 won
wins = 1
print(f"{wins}승")
```

Hint: add to the current value with `wins = wins + 1`.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code

Read this working tournament closely. Then answer the questions below in your own words.

```

import random

TYPE_ADVANTAGE = {"물": "불", "불": "풀", "풀": "전기", "전기": "물"}

def calculate_score(attacker, defender):
    base = attacker["hp"] + attacker["attack"]
    if TYPE_ADVANTAGE.get(attacker["type"]) == defender["type"]:
        base = int(base * 1.2)
        print(f" ⚡ {attacker['name']}의 {attacker['type']} 타입이 효과적입니다!")
    return base

def battle_round(my_p, enemy_p):
    my_score = calculate_score(my_p, enemy_p)
    enemy_score = calculate_score(enemy_p, my_p)
    if my_score > enemy_score:
        return "win"
    elif my_score < enemy_score:
        return "lose"
    else:
        return "draw"

def tournament(my_pokemon, pokedex):
    wins = 0
    losses = 0
    print(f"\n=== 토너먼트 시작! 내 포켓몬: {my_pokemon['name']} ===")
    while wins < 2 and losses < 2:
        enemy = random.choice(pokedex)
        print(f"\n🔴 {my_pokemon['name']} vs {enemy['name']}")
        result = battle_round(my_pokemon, enemy)
        if result == "win":
            wins = wins + 1
            print(f" → 승리! ({wins}승 {losses}패)")
        elif result == "lose":
            losses = losses + 1
            print(f" → 패배... ({wins}승 {losses}패)")
        else:
            print(" → 무승부, 재경기!")
    if wins == 2:
        print(f"\n🏆 토너먼트 우승! {my_pokemon['name']} 최강!")
    else:
        print(f"\n😞 토너먼트 패배. 다음 기회에...")

pokedex = [
    {"name": "피카츄", "type": "전기", "hp": 35, "attack": 55},
    {"name": "이상해씨", "type": "풀", "hp": 45, "attack": 49},

```

```
    {"name": "꼬부기", "type": "물", "hp": 44, "attack": 48},  
    {"name": "파이리", "type": "불", "hp": 39, "attack": 52},  
]  
  
tournament(pokedex[0], pokedex)
```

In `calculate_score`, what makes `base` get multiplied by `1.2`?

The `while` loop says `while wins < 2 and losses < 2`. When does this loop stop?

Why does `calculate_score` use `TYPE_ADVANTAGE.get(...)` instead of `TYPE_ADVANTAGE[...]`?

Where does the enemy Pokémon come from each round? Which line picks it?

After the loop ends, how does the code decide whether to print "우승" or "패배"?

5 · Explain Your Lesson Code 🎥

Make a short phone video where you explain the tournament code **you wrote in today's lesson**. You may show it running. Teach it like the viewer is new. Try to use: **tournament, while loop, type advantage, get, best of three**.

1. Show the code you wrote and say what it does overall.
2. Point to the `while wins < 2 and losses < 2` line and explain it.
3. Show or describe a round where the type bonus kicks in.
4. Explain why `.get` keeps the program from crashing.

Write what you will say in your video. Plan it here before you record.

Submit ☒

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.